

Assignment 1: Data Lakehouse

Alizée Marchand - 25761139

Big Data Engineering- Spring 2026



Table of Contents

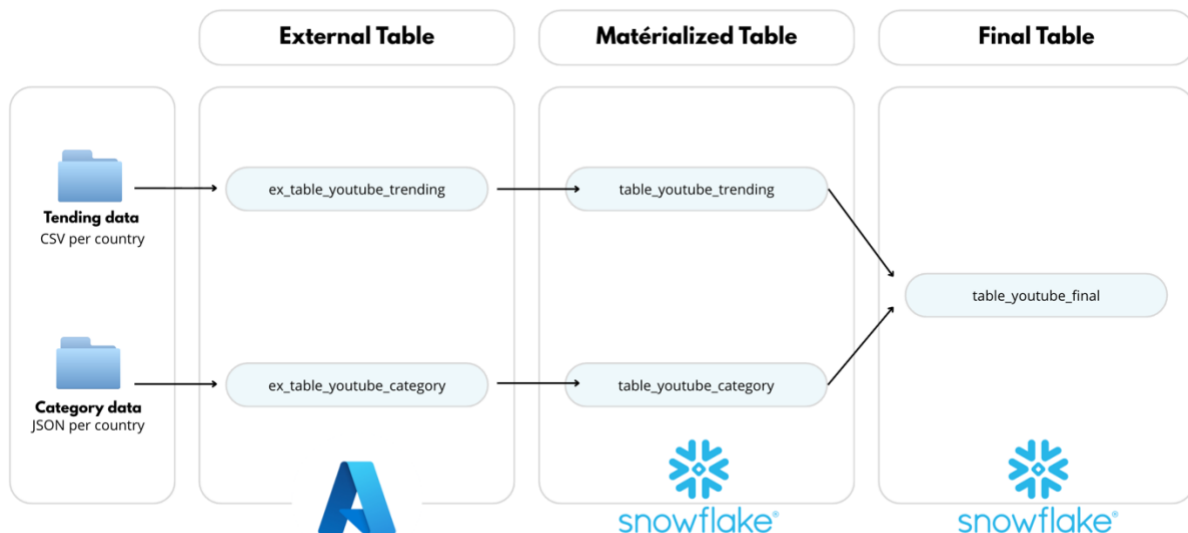
Project Overview	2
Part 1 — Data Ingestion	2
Part 2 — Data Cleaning	4
Part 3 — Data Analysis	5
Part 4 — Business Question	7
Business reflection	7
General Investigation	7
Which Categories Are Realistic Targets	7
Is This a Long-Term Trend or a Short-Term Buzz	7
Comparing a Full Year to a Partial Year	8
Deep Category Analysis	9
Channel Diversity per Category	9
Channel Concentration in Each Category	9
People & Blogs Performance Across Countries	10
Popularity Analysis	11
Video Turnover per Category	11
Audience Size per Category	11
Will This Strategy Work in Every Country	13
The France Exception	13
Final Recommendation	14
APPENDIX	15
APPENDIX 1- Data Catalog	15

Project Overview

For this assignment, I built a small data lakehouse pipeline in Snowflake to ingest, clean, and analyse the YouTube Trending Video dataset, sourced from Azure Blob Storage. The dataset covers 10 countries (BR, CA, DE, FR, GB, IN, JP, KR, MX, US) and combines two file types per country: a CSV of daily trending videos (video metadata and engagement metrics) and a JSON file mapping category IDs to category names.

The goal was to move this raw, per-country data into a single, clean, queryable table in Snowflake, then use it to answer both structured analytical questions and an open-ended business question. The project is split into four parts: ingesting the raw files as external tables and consolidating them into one final table (Part 1), identifying and fixing data quality issues in that table (Part 2), running exploratory analysis to answer specific questions about trending patterns (Part 3), and using the cleaned dataset to make a data-driven recommendation for launching a new YouTube channel (Part 4).

Figure 1: Data Flow Chart



Part 1 — Data Ingestion

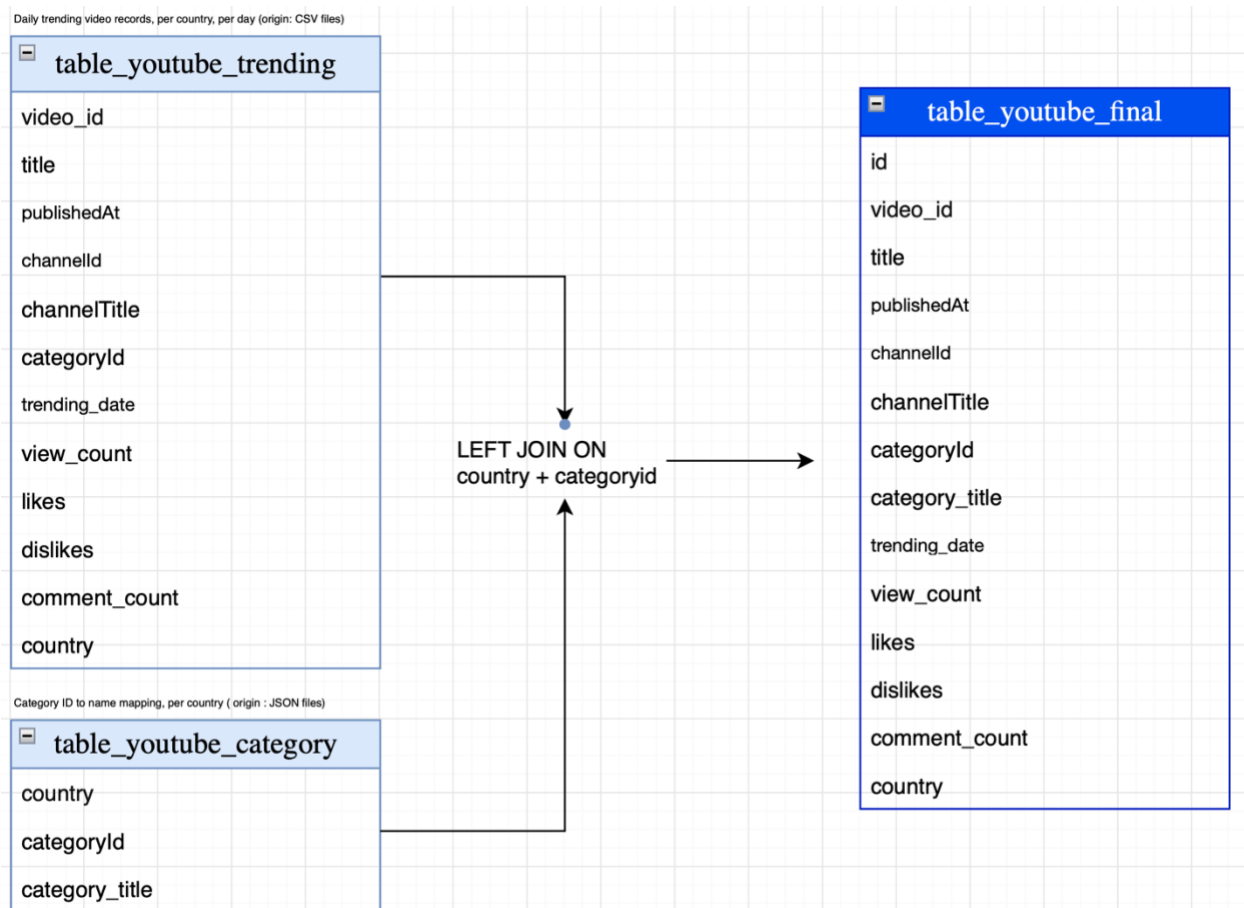
stage_assignment points to the Azure container using a SAS token, so nothing actually gets copied at this stage, it's just a pointer. From there, two external tables read directly off the stage: **ex_table_youtube_trending**, where I typed each of the 11 CSV columns explicitly, and **ex_table_youtube_category**, which just reads the JSON as is since its nested structure can't really be typed like a flat CSV.

These two get turned into materialized tables next. For the trending data, I pulled the country out of the filename itself (BR_youtube_trending_data.csv becomes BR) using split_part. For the category data, I used

a lateral flatten to unfold the nested items array in each JSON into one row per category, extracting country the same way.

Finally, table_youtube_final joins both on country and categoryid with a left join, so no trending video gets dropped even without a category match, and adds a UUID per row that I later use in Part 2 to remove duplicates. Final row count is 2,667,041, matching the brief exactly.

Figure 2: Data Integration Model with Draw.io



Part 2 — Data Cleaning

Duplicate Category Label in the Reference Data

A category should normally show up once per country, so at most 10 rows across 10 countries. I looked for any category with more rows than distinct countries, since that would mean it showed up twice somewhere. Only Comedy matched: all 10 countries had it listed twice, once under categoryid 23 and once under 34.

Category Present in a Single Country

Same logic here, just checking which category was tied to only one country: Nonprofits & Activism.

Identifying the Missing Category

table_youtube_final comes from a left join on country and categoryid, so if a categoryid was missing from a country's category file, category_title ends up NULL for those rows. Before fixing anything I checked how big the issue actually was: one categoryid, 29, affecting 1,563 rows. Looking it up on YouTube's official category list, categoryid 29 is Nonprofits & Activism, the same one from the step above, which also explains why it's missing in some countries, it's just not a category every country has.

Correcting the Missing Category Values

I updated table_youtube_final to fill category_title with Nonprofits & Activism wherever it was NULL. Rechecking the total for that category afterward gave 1,683, which lines up with the 1,563 rows I fixed plus the ones that already had it.

Video Missing a Channel Title

Filtered for NULL channeltitle and got a single video back, so this was just a one-off, not something affecting a bunch of rows.

Removing Corrupted Video IDs

Some rows had video_id equal to #NAME?, which is the error Excel throws when a cell reference breaks, probably from how the original files got put together. No real video_id means nothing useful to keep, so I deleted these, 32,081 rows gone.

Detecting Duplicate Trending Records

Some rows had the same video_id, country and trending_date but different metrics, meaning the same video got recorded more than once. The brief says to keep whichever row has the highest view_count. So I grouped by video_id, country and trending_date, sorted each group by view_count descending, and used ROW_NUMBER() to flag everything after the first row in each group as a duplicate. That gave 37,466 rows, saved into table_youtube_duplicates. As a sanity check I also just counted, directly in table_youtube_final, how many rows shared those same three fields: 74,932, exactly double 37,466. So every duplicate group had exactly 2 rows, one to keep and one to toss, which confirmed the logic was doing what it should.

Removing the Duplicate Records

Deleted those 37,466 rows from table_youtube_final, matching on id, the UUID from Part 1.

Part 3 — Data Analysis

Top 3 Most Viewed Gaming Videos per Country on 2024-04-01

For 2024-04-01, I found the 3 most viewed Gaming videos in each country using ROW_NUMBER(). The same video, DAGGER DUCHESS - New Tower Troop!, was the number 1 most viewed Gaming video in 8 out of 10 countries that day, showing how one trending video can dominate across very different regions at once.

TableChartPivot

30 rows607ms

	COUNTRY	TITLE	VIEW_COUNT	RANK
	BR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	4923026	1
	CA	If my viewers break my secret rule, I ban them	2988844	2
	+8 more	+16 more		3
1	BR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	4923026	1
2	BR	IShowSpeed x MC Kevin O Chris - Amar de (Official Music Video)	2971782	2
3	BR	Confrontation - The Skibidi Saga 05	2323375	3
4	CA	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	4923026	1
5	CA	If my viewers break my secret rule, I ban them	2988844	2
6	CA	Confrontation - The Skibidi Saga 05	2323375	3
7	DE	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	4923026	1
8	DE	If my viewers break my secret rule, I ban them	2988844	2
9	DE	Season 3 Warzone Launch Trailer - Rebirth Island Call of Duty: Warzone	2311131	3
10	FR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	4923026	1
11	FR	Season 3 Warzone Launch Trailer - Rebirth Island Call of Duty: Warzone	2311131	2
12	FR	Clove Official Gameplay Reveal // VALORANT	2043592	3
13	GB	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	4923026	1
14	GB	If my viewers break my secret rule, I ban them	2988844	2

Counting Videos Related to BTS

I counted distinct videos with "BTS" in the title per country, using COUNT(DISTINCT video_id) so repeated trending dates don't inflate the count. South Korea has by far the most, with 468, which makes sense given BTS is a Korean group.

TableChartPivot

10 rows

	COUNTRY	NB_VIDEOS
	BR	167
	CA	262
	+8 more	468
1	KR	468
2	IN	288
3	US	268
4	CA	262
5	MX	254
6	JP	251
7	DE	242
8	GB	223
9	BR	186
10	FR	167

Most Viewed Video per Country and Month in 2024

For each country and month in 2024, I found the most viewed video and its likes_ratio, likes as a percentage of view_count. MrBeast's videos dominate January to March across most countries. April stands out: the video Discord Loot Boxes are here. reached over 1.4 billion views in several countries, but its likes_ratio is only 0.00 to 0.01%, far below the usual 1-6% range.

	COUNTRY	YEAR_MONTH	TITLE	CHANNELTITLE	CATEGORY_TITLE	VIEW_COUNT	LIKES_RATIO
	BR 10.0% CA 10.0% +8 more	1/1/2024 4/1/2024	Face Your Biggest Fear To Win \$800,000 27.5% I Survived 7 Days In An Abandoned City 22.5% +8 more	MrBeast 67.5% Discord 10.0% +4 more	Entertainment 82.5% Gaming 12.5% +1 more	85458562 14076436...	0 6.72
1	BR	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	139504939	3.20
2	CA	2024-01-01	Still Here Season 2024 Cinematic - League of Legends	League of Legends	Gaming	104159411	1.68
3	DE	2024-01-01	Still Here Season 2024 Cinematic - League of Legends	League of Legends	Gaming	104159411	1.68
4	FR	2024-01-01	Still Here Season 2024 Cinematic - League of Legends	League of Legends	Gaming	104159411	1.68
5	GB	2024-01-01	Still Here Season 2024 Cinematic - League of Legends	League of Legends	Gaming	104159411	1.68
6	IN	2024-01-01	Protect \$500,000 Keep It!	MrBeast	Entertainment	85458562	4.21
7	JP	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	137639799	3.22
8	KR	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	143955997	3.15
9	MX	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	137639799	3.22
10	US	2024-01-01	Grand Theft Auto VI Trailer 1	Rockstar Games	Gaming	166323421	6.72
11	BR	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	126846652	3.72
12	CA	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	119170728	3.72
13	DE	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	114978689	3.72
14	FR	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	114978689	3.72
15	GB	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	114978689	3.72
16	IN	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	109797680	3.80
17	JP	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	130451673	3.48

Top Category per Country in 2022

For each country, I found the top category by distinct video count in 2022, and what share of the country's total trending videos it represents. Entertainment is the top category in 8 out of 10 countries, and Gaming in the remaining 2, Canada and the United States. India has the highest concentration, Entertainment alone makes up 37.53% of its trending videos, while the United Kingdom has the lowest at 20.94%.

	COUNTRY	CATEGORY_TITLE	TOTAL_CATEGORY_VIDEO	TOTAL_COUNTRY_VIDEO	PERCENTAGE
	BR 10.0% CA 10.0% +8 more	Entertainment 80.0% Gaming 20.0%	2017 8566	6916 22821	20.94 37.53
1	BR	Entertainment	2715	11316	23.99
2	DE	Entertainment	3373	13660	24.69
3	FR	Entertainment	3663	15305	23.93
4	GB	Entertainment	2588	12355	20.94
5	IN	Entertainment	8566	22821	37.53
6	JP	Entertainment	2655	8305	31.96
7	KR	Entertainment	2437	6916	35.23
8	MX	Entertainment	2017	8471	23.81
9	CA	Gaming	3376	14772	22.85
10	US	Gaming	3138	13946	22.50

Channel With the Most Videos

I counted, for each channeltitle, how many distinct videos it produced across the whole dataset, meaning all countries and all dates combined, then kept only the channel with the highest count. The channel with the most distinct videos is Vijay Television, with 2,049 different videos.

	CHANNELTITLE	NB_VIDEOS
1	Vijay Television	2049

Part 4 — Business Question

Business reflection

Before running any query, it's worth thinking through why this question is trickier than picking whatever category has the most trending videos.

First, we only see videos that already trended, not total uploads per category, so a high trending count could just mean a category gets flooded with uploads, not that it's easy to break into. I don't have upload volume to calculate a real success rate, so this stays a limitation rather than something the analysis can fully resolve.

Second, and more importantly, raw volume isn't the same as real opportunity: if a category's trending videos mostly come from two or three already-massive channels, there is little room left for a new creator, no matter how big the category looks. Volume is only a first filter, the number of distinct channels reaching trending matters just as much.

Third, a category where many different videos cycle through trending, high turnover, offers more openings than one where the same handful of videos sit at the top for weeks.

Fourth, trending is calculated per country, not globally, so whether the strategy works in every country is central to the question, not a side check. No single category may work everywhere, and "it depends on the market" can be the honest answer.

Fifth, the dataset spans several years, but the brief asks about launching tomorrow. A category that dominated in the past could already be declining now, so the trend over time matters more than an average across the whole period.

Finally, category_title itself is not perfectly clean, part 2 already found mapping issues, so any recommendation based on it inherits some of that uncertainty.

General Investigation

Which Categories Are Realistic Targets

Since Music and Entertainment are excluded from the business question, I first looked at video volume for all the remaining categories, to see which ones actually have a meaningful presence in the trending lists. There is a clear gap after the top four categories: Sports with 43,323 videos, People & Blogs with 42,876, Gaming with 41,615, and Comedy with 18,605. Every other category falls far behind these four, so I focused the rest of the analysis only on these four candidates.

Is This a Long-Term Trend or a Short-Term Buzz

To find out if the top four categories are staying popular over time or if they just had one busy year, I looked at the number of videos per category for every year in the dataset. Since the dataset only partially

covers 2020, starting August 12, and 2024, ending April 15, I first checked how many days of data each year actually had, to make sure any year-to-year comparison only used complete years. This confirmed that 2021, 2022 and 2023 are the only full years, with 353, 364 and 362 days covered respectively, so I focused the trend comparison on those three years only. Using LAG(), I compared each year to the previous one to calculate a percentage change. Sports is the only category that grew from 2022 to 2023, up 12.03%. The other three categories all declined that year: Gaming fell 16.66%, People & Blogs fell 23.45%, and Comedy fell the most, 33.70%, after already slipping slightly the year before. This shows that these categories are not a short-term buzz, since they have consistently produced thousands of trending videos across three full years, but their popularity is not equally stable: Sports looks the most resilient, while Comedy is the one losing ground the fastest.



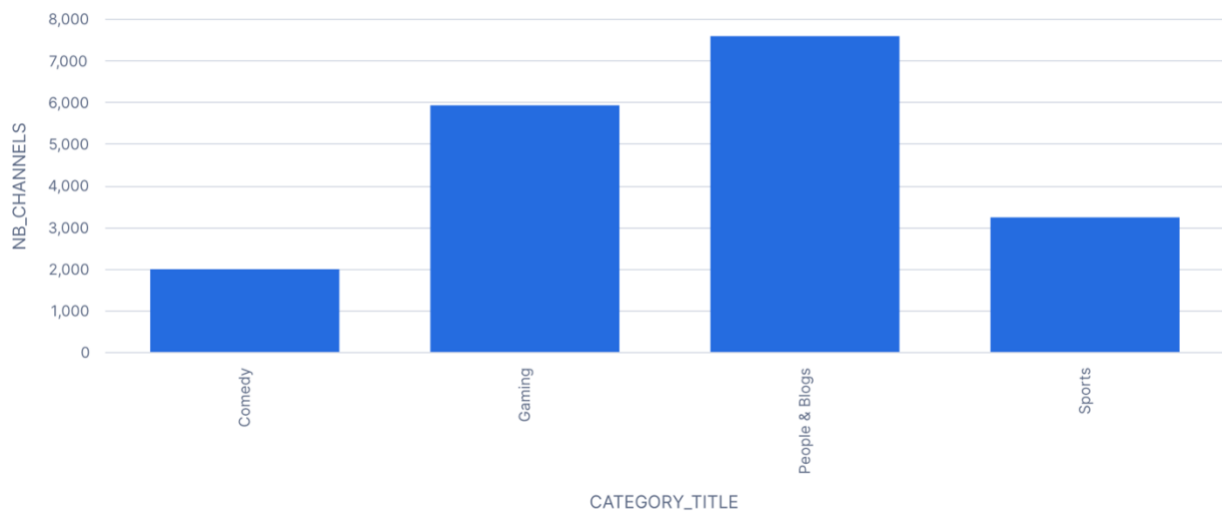
Comparing a Full Year to a Partial Year

Since 2024 only has data up to April 15, comparing its raw video count to 2023 directly would be misleading, a category could look like it's declining just because the year isn't finished yet. To compare fairly, I normalized both years by dividing the number of videos by the number of days covered, giving a videos-per-day rate for each category, then calculated the percentage change between 2023 and 2024 using this rate instead of the raw count. Every one of the four categories shows a decline in its daily rate, but not by the same amount. Gaming has the steepest drop, down 17.14%, while People & Blogs is the most stable, down only 2.20%. Sports declined 6.86% and Comedy 10.17%. This confirms what the yearly trend already suggested: Gaming's popularity is dropping the fastest among the four categories, while People & Blogs is holding up the best.

Deep Category Analysis

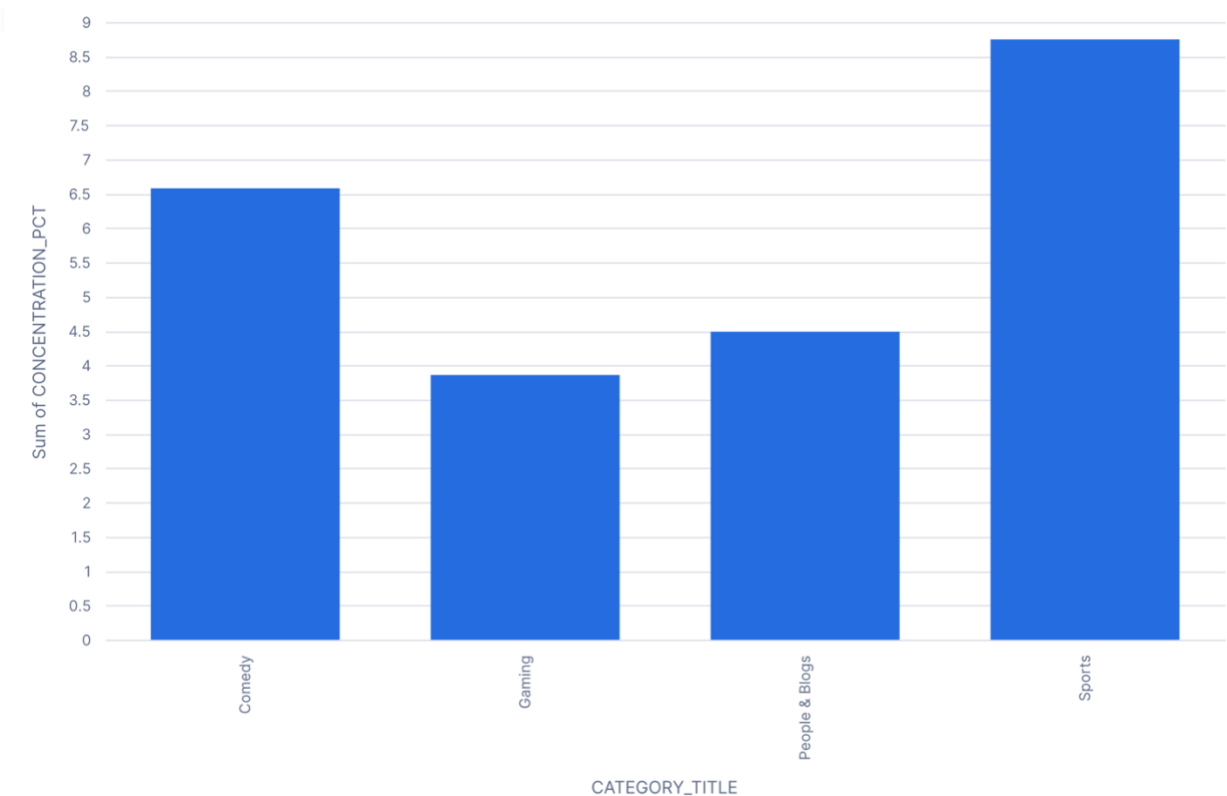
Channel Diversity per Category

For each of the four categories, I counted the total number of distinct videos and the number of distinct channels that produced them, to get a first sense of how spread out or concentrated each category is. People & Blogs has clearly the most channels, 7,583, even though it has slightly fewer total videos than Sports. Sports, on the other hand, has the fewest channels of the four, 3,232, despite having the highest video count.



Channel Concentration in Each Category

To see whether a small number of channels dominate each category, I calculated what share of each category's total videos came from just its top 5 channels. Sports has the highest concentration, 8.75%. Gaming has the lowest concentration, 3.86%. Comedy sits at 6.58% and People & Blogs at 4.49%.



People & Blogs Performance Across Countries

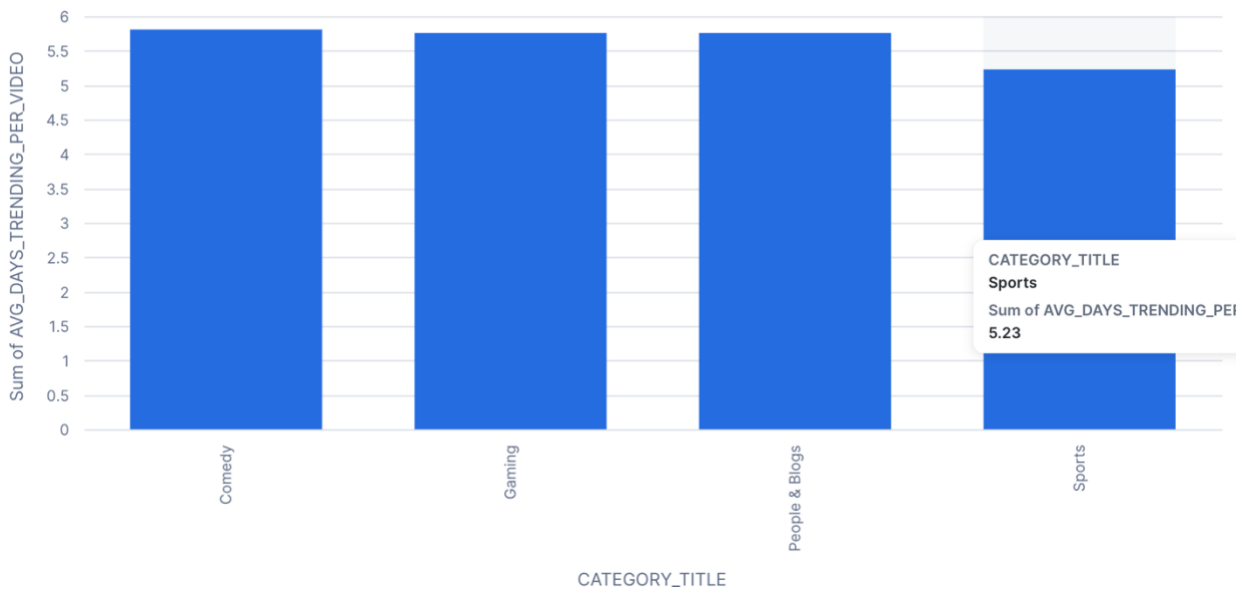
Since People & Blogs stood out as accessible and stable, I checked whether it performs well everywhere or just on average. It ranks in the top 3 in 9 out of the 10 countries: number 1 in South Korea and India, number 2 in Mexico and Japan, and number 3 in Canada, the United States, Brazil, Germany and the United Kingdom. France is the only exception, where it only ranks 4th.

COUNTRY		CATEGORY_TITLE	NB_VIDEOS	RANK_IN_COUNTRY
BR	10.0%	People & Blogs	2870	1
CA	10.0%	People & Blogs	4343	3
+8 more				
1 KR		People & Blogs	4379	1
2 IN		People & Blogs	12580	1
3 MX		People & Blogs	3927	2
4 JP		People & Blogs	3720	2
5 CA		People & Blogs	4343	3
6 US		People & Blogs	3991	3
7 BR		People & Blogs	4536	3
8 DE		People & Blogs	5604	3
9 GB		People & Blogs	4172	3
10 FR		People & Blogs	2870	4

Popularity Analysis

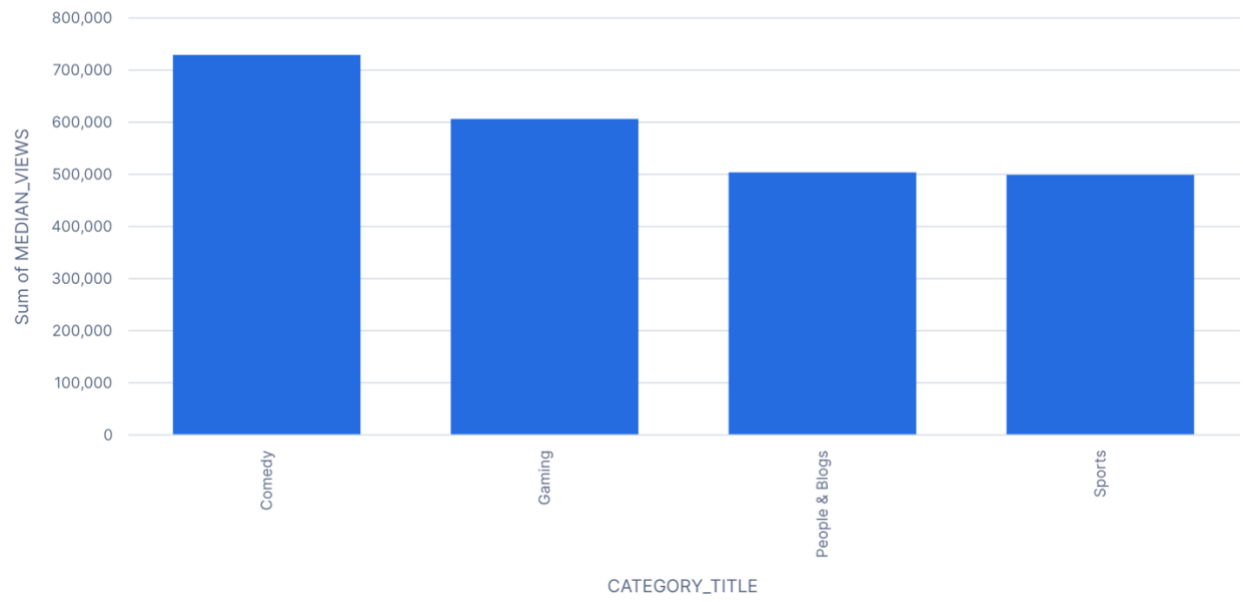
Video Turnover per Category

For each video, in each country, I counted how many different days it stayed in the trending list, then averaged this per category. A lower average means more turnover, more opportunities for new videos to appear. Sports has the fastest turnover, 5.23 days on average. Comedy has the slowest, 5.81 days. Gaming and People & Blogs are almost identical, at 5.76 days each.



Audience Size per Category

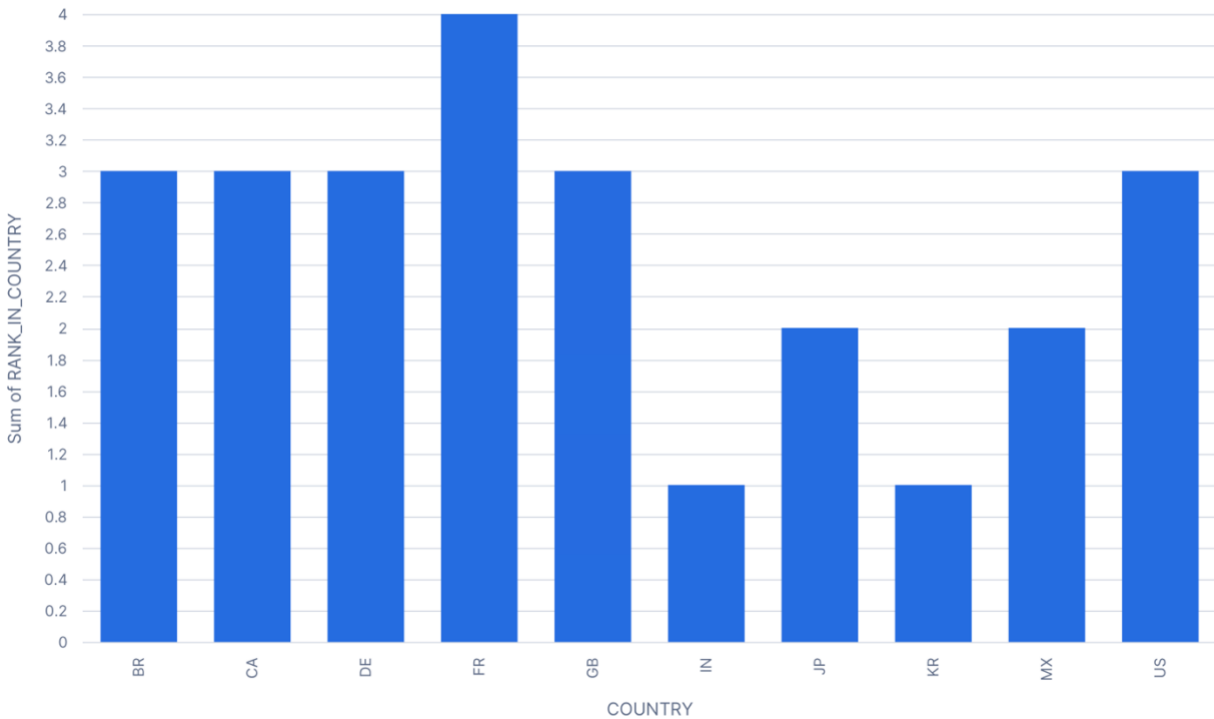
I checked, for each category, the average and median view_count, focusing mainly on the median since a few viral videos can pull the average up. Comedy has the highest typical view count, 727,947, followed by Gaming at 605,051, then People & Blogs at 502,560, and Sports the lowest at 497,879.



Will This Strategy Work in Every Country

The France Exception

Since People & Blogs ranked outside the top 3 only in France, I checked directly what does rank in the top 3 there. In France, the leading categories are Gaming with 7,788 videos, Sports with 6,828, and Comedy with 4,116.



Final Recommendation

Looking at all the signals together, the category I would recommend is People & Blogs.

Sports has the highest volume and was the only category growing over the three full years studied, but its low channel diversity and high concentration, shown in Deep Category Analysis, make it the hardest to break into: a new channel would be competing against an already established group of creators.

Gaming is the most open category by concentration, but also the one declining fastest recently. Entering an accessible category that is actively shrinking is a real risk.

People & Blogs performs strongly across most signals at once: real scale, the most channels and the second lowest concentration, the most stable trend into 2024, and consistent top-3 performance across 9 of the 10 countries.

Its main weakness is audience size: as shown in Audience Size per Category, it has the third highest median view count of the four, behind Comedy and Gaming, so individual videos tend to attract somewhat fewer views even though the category is easier to enter and more stable.

My last personal recommendation for a new channel would be not to focus only on one category but try to create different content to get the best of each category. It also helps to check the videos of established group of creators to better understand their knowhow for creating a content that will make sure to be liked.

APPENDIX

APPENDIX 1- Data Catalog

Final Data Layer

The Final Data Layer represents the query-ready data used to answer all business questions in Part 2, 3 and 4. Unlike a traditional star schema with separate fact and dimension tables, this project uses a One Big Table design: the trending and category data are joined once during ingestion into a single denormalized table, so no further joins are needed at query time.

1. table_youtube_trending

Purpose: Stores one row per video per country per trending day, extracted from the 10 CSV files provided (one per country).

Column Name	Data Type	Description
video_id	VARCHAR	Unique identifier of the trending video, as assigned by YouTube.
title	VARCHAR	Title of the video as displayed on YouTube.
publishedat	TIMESTAMP_NTZ	Date and time the video was originally published on YouTube.
channelid	VARCHAR	Unique identifier of the YouTube channel that published the video.
channeltitle	VARCHAR	Name of the YouTube channel that published the video.
categoryid	INT	Numeric identifier of the video's category, used to join with the category data.
trending_date	DATE	Date on which the video appeared in the trending list.
view_count	INT	Total number of views recorded for the video on the trending date.
likes	INT	Total number of likes recorded for the video on the trending date.
dislikes	INT	Total number of dislikes recorded for the video on the trending date.
comment_count	INT	Total number of comments recorded for the video on the trending date.

country	VARCHAR	Two-letter country code identifying the trending list the video appeared in, extracted from the source filename.
---------	---------	--

2. table_youtube_category

Purpose: Stores the category ID to category name mapping for each country, extracted from the 10 JSON files provided (one per country).

Column Name	Data Type	Description
country	VARCHAR	Two-letter country code identifying which country's category list this row belongs to, extracted from the source filename.
categoryid	INT	Numeric identifier of the video category, as defined by YouTube.
category_title	VARCHAR	Human-readable name of the video category (e.g., Gaming, Music, Sports).

3. table_youtube_final

Purpose: Stores the full, query-ready dataset combining trending and category data for every country. This is the single table used to answer every business question in Part 2, 3 and 4.

Column Name	Data Type	Description
id	VARCHAR	Surrogate key uniquely identifying each row, generated with UUID_STRING().
video_id	VARCHAR	Unique identifier of the trending video, as assigned by YouTube.
title	VARCHAR	Title of the video.
publishedat	TIMESTAMP_NTZ	Date and time the video was originally published.
channelid	VARCHAR	Unique identifier of the YouTube channel.
channeltitle	VARCHAR	Name of the YouTube channel.
categoryid	INT	Numeric identifier of the video's category.

category_title	VARCHAR	Human-readable name of the video category, joined in from table_youtube_category.
trending_date	DATE	Date on which the video appeared in the trending list.
view_count	INT	Total number of views recorded on the trending date.
likes	INT	Total number of likes recorded on the trending date.
dislikes	INT	Total number of dislikes recorded on the trending date.
comment_count	INT	Total number of comments recorded on the trending date.
country	VARCHAR	Two-letter country code identifying the trending list the video appeared in.

Final Data Layer Architecture

This project uses a One Big Table design rather than a star schema:

- ◇ table_youtube_trending and table_youtube_category are combined once, at ingestion time, through a LEFT JOIN on country and categoryid.
- ◇ The result, table_youtube_final, is a single denormalized table containing every attribute needed for analysis, avoiding repeated joins across Part 2, 3 and 4.
- ◇ This design fits the scope of the assignment, which asks for one clean, analysis-ready table rather than a multi-table dimensional model.